Stabilizing Two-Stage Corrections for Multinomial Selection through Regularization

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Abstract

This note examines a regularized two-stage correction procedure for selection bias in high-dimensional multinomial choice models. The first stage estimates a penalized multinomial logit to obtain stable choice probabilities when covariates are numerous or collinear. The second stage augments the outcome equation with polynomial terms and a Lee (1983) type correction term derived from these probabilities, while applying regularization to stabilize estimation. Monte Carlo evidence indicates that regularization in both stages reduces finite-sample bias and mean squared error in the target linear coefficient, particularly under sparsity. An empirical illustration based on a transportation mode choice dataset demonstrates that the approach mitigates endogeneity-induced bias in practice. The framework provides an accessible and computationally stable extension of classical selection correction methods to modern, high-dimensional empirical applications.

- *Keywords*: Sample Selection, Multinomial Logit, High-Dimensional Estimation, Regularization, LASSO, Ridge Regression
- *JEL Classification*: C13, C24, C25, C52

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1 Introduction

Econometric analyses frequently encounter challenges related to sample selection, wherein the outcome variable of interest is observed only for a non-random subset of the population. Ignoring such selection typically results in biased and inconsistent parameter estimates. The seminal work of Heckman (1979) introduced a two-stage correction for binary selection models, which remains a cornerstone of applied microeconometrics. This framework was subsequently generalized to accommodate multiple discrete selection alternatives by Lee (1983) and Dubin and McFadden (1984), enabling its application to contexts such as occupational choice, brand choice, and transportation mode selection.

In contemporary empirical research, analysts increasingly face datasets containing a large number of potential predictors (p) relative to the number of observations (n). Such high-dimensional environments pose substantial difficulties for traditional selection correction methods. First, the selection equation—typically specified as a multinomial logit or probit model—becomes unstable or even inestimable by conventional maximum likelihood when p is large, particularly under multicollinearity. Second, the outcome equation may involve complex nonlinear relationships or heteroskedastic, non-normal disturbances that are difficult to approximate without flexible functional forms, which in turn heighten the risk of overfitting. Unregularized two-stage correction procedures are known to perform poorly under these conditions because estimation noise from the first stage tends to propagate into the second stage, producing unreliable results (Bourguignon et al., 2007).

This note proposes a regularized extension of the classical two-stage correction framework for multinomial selection. The core idea is to stabilize both stages of estimation through penalized regression methods such as LASSO (Tibshirani, 1996) and ridge regression. In the first stage, a penalized multinomial logit in the style of Friedman et al. (2010) is employed to estimate choice probabilities. This approach performs variable selection and coefficient shrinkage, thereby yielding stable predicted probabilities even when covariates are numerous and highly correlated. In the second stage, the outcome equation is augmented with a Lee-type correction term computed from the regularized first-stage probabilities. To accommodate potential nonlinearities, low-order polynomial terms of the main regressor are included. Regularization is again applied in this stage, with the critical modification that the selection correction term is left unpenalized to preserve its bias-correction function. This design aligns with principles from high-dimensional inference and double/debiased machine learning, which advocate regularization for nuisance parameter estimation while safeguarding the parameter of interest (Belloni et al., 2013; Chernozhukov et al., 2018).

The contribution of this note is not to propose a new estimator in a theoretical sense, but rather to develop and evaluate a specific, practical combination of existing high-dimensional techniques to address a common problem in applied econometrics. While each component—penalized multinomial logit and regularized linear modeling—is well established, their integration within a multinomial selection framework provides a coherent and robust empirical strategy that has not been systematically examined. The approach is deliberately pragmatic, targeting applied researchers operating in data-rich environments where conventional methods are unreliable. Its performance is assessed through extensive Monte Carlo simulations under conditions of strong endogeneity, sparsity, multicollinearity, and non-standard errors. An empirical illustration based on a transportation mode choice dataset further demonstrates the method's ability to correct large biases that would otherwise render standard estimators ineffective. The note remains focused on finite-sample behavior rather than asymptotic theory, consistent with its methodological scope. Alternative approaches, such as the semiparametric correction methods discussed in Dahl (2002), offer conceptually appealing solutions but tend to involve more complex implementation, making the proposed regularized two-stage procedure an attractive and accessible alternative.

2 Model and Selection Correction

Consider a setting with J discrete alternatives, where agent i selects the alternative that maximizes their latent utility. The latent utility of agent i for alternative j is specified as a linear-in-parameters model:

$$u_{ij}^* = Z_i\gamma_j + \eta_{ij}, \quad j = 1, 2, ..., J, \tag{1}$$

where Z_i denotes a $1 \times p$ vector of covariates determining the choice, γ_j is a corresponding $p \times 1$ parameter vector for alternative j, and η_{ij} represents unobserved disturbances. The chosen alternative, D_i, satisfies $D_i = \arg\max_{j \in \{1,...,J\}} u_{ij}^*$. Under the assumption that η_{ij} are independently and identically distributed with a Type I extreme value (Gumbel) distribution, the model reduces to the standard multinomial logit model (McFadden, 1973). For identification, the coefficients of one alternative are normalized to zero, typically $\gamma_J = 0$. The probability that agent i selects alternative j is therefore given by:

$$P_{ij} = P(D_i = j|Z_i) = \frac{\exp(Z_i\gamma_j)}{\sum_{k=1}^{J}\exp(Z_i\gamma_k)}. \tag{2}$$

The outcome of interest, y_i, is observed only when a particular alternative is chosen, for instance, alternative 1 ($D_i = 1$). The outcome equation is specified as a potentially nonlinear regression model:

$$y_i = \beta X_i + g(X_i) + u_i, \quad \text{observed only if } D_i = 1, \tag{3}$$

where X_i denotes the main regressor of interest, β is the coefficient associated with the linear effect of X_i, and $g(X_i)$ captures other nonlinear components (e.g., polynomial or spline terms). The disturbance term u_i may be heteroskedastic and non-normally distributed. Selection bias arises when the unobservables in the selection equation, η_{ij}, are correlated with those in the outcome equation, u_i. Such correlation is plausible in many economic applications; for example, unobserved ability may jointly affect an individual's occupational choice and their resulting wage within that occupation.

To address this bias, a two-stage correction procedure is typically employed. Following the framework of (Lee, 1983), the conditional expectation of the outcome error

u_i given that alternative 1 is chosen can be expressed as a function of the corresponding choice probability P_{i1}. This leads to the augmented outcome regression:

$$E[y_i|D_i = 1, X_i, Z_i] = \beta X_i + g(X_i) + E[u_i|D_i = 1, X_i, Z_i]. \tag{4}$$

(Lee, 1983) shows that, under specific distributional assumptions—particularly that η_{ij} and u_i are linked through a bivariate normal structure involving a transformed selection error term—the conditional expectation $E[u_i|D_i = 1]$ can be written as $\sigma_u \rho \lambda_i$, where σ_u denotes the standard deviation of u_i, ρ is the correlation coefficient, and λ_i is a control function. The most common form of this correction term, generalizing the inverse Mills ratio from the binary case, is:

$$\lambda_i = -\frac{\phi\big(\Phi^{-1}(P_{i1})\big)}{P_{i1}}, \tag{5}$$

where $\phi(\cdot)$ and $\Phi^{-1}(\cdot)$ represent the probability density and quantile functions of the standard normal distribution, respectively. This transformation effectively converts the non-randomness of the selected subsample into a form that can be addressed through an omitted-variable correction. Including λ_i as an additional regressor in the outcome equation allows the model to absorb selection-related dependence, facilitating consistent estimation of β.

Although alternative correction formulations exist—such as the Dubin–McFadden approach, which incorporates all choice probabilities—the single-index correction proposed by Lee (1983) is adopted here due to its parsimony and its close analogy to the well-known Heckman (1979) model(Heckman, 1979). This structure provides a clear foundation for integrating regularization in both estimation stages while maintaining interpretability and computational tractability.

3 Estimation Procedure

The proposed procedure extends the traditional two-stage framework to high-dimensional settings by incorporating regularization at both stages.

Step 1: Regularized First-Stage Estimation

The first stage estimates the multinomial logit model to obtain the choice probabilities P_{i1}. When the number of covariates p is large, conventional maximum likelihood estimation becomes ill-posed or unstable. To address this issue, a penalized multinomial logistic regression is employed using the implementation of Friedman et al. (2010), as available through the `cv.glmnet` function in R. The estimator solves the following penalized log-likelihood optimization problem for either the LASSO ($\alpha = 1$) or ridge ($\alpha = 0$) penalty:

$$\max_{\{\gamma_j\}_{j=1}^{J-1}} \left[\sum_{i=1}^{n} \log P(D_i|Z_i, \{\gamma_j\}) \right] - \lambda_1 \sum_{j=1}^{J-1} \big((1-\alpha)\|\gamma_j\|_2^2/2 + \alpha\|\gamma_j\|_1\big). \tag{6}$$

The penalty parameter λ_1 is selected via k-fold cross-validation to minimize prediction error. This step yields a set of stable and possibly sparse parameter estimates $\{\hat{\gamma}_j\}$, from which predicted probabilities $\hat{P}_{i1}$ are computed. The estimated probabilities are then used to construct the Lee-type correction term $\hat{\lambda}_i$ for each observation i using equation (5).

Step 2: Regularized Second-Stage Estimation

In the second stage, the outcome equation is estimated using only the subsample for which $D_i = 1$. The specification includes the estimated correction term and low-order polynomial terms of X_i to flexibly approximate potential nonlinearities:

$$y_i = \beta_0 + \beta X_i + \delta_2 X_i^2 + \delta_3 X_i^3 + \theta\hat{\lambda}_i + \epsilon_i. \tag{7}$$

Given possible multicollinearity among regressors (X_i, its powers, and $\hat{\lambda}_i$) and the high-dimensional nature of the model, a penalized linear regression is again employed. A key feature of this step is the use of differentiated penalty factors. The selection correction term $\hat{\lambda}_i$ is essential for bias adjustment and should not be penalized; accordingly, its penalty factor is set to zero (`penalty.factor=0`). The remaining regressors—particularly the polynomial terms—are penalized to enhance stability and control variance. The penalty parameter λ_2 is determined by cross-validation. The final estimate of the parameter of interest, $\hat{\beta}$, corresponds to the coefficient on the linear term X_i. Although formal inference on $\hat{\beta}$ is beyond the scope of this note, standard errors can be obtained through bootstrap resampling. The primary objective of this approach is finite-sample bias and variance reduction rather than asymptotic inference.

4 Results

4.1 Simulation Evidence

Table 1 reports the Monte Carlo simulation results for three representative dimensionalities: $p = 5$, $p = 100$, and $p = 1000$. The results reveal consistent patterns across all settings. Both the Naive OLS and unregularized Lee estimators perform poorly, exhibiting large bias (approximately 142–146) and extremely high mean squared error (MSE) values (exceeding 20,000). These outcomes are expected, as neither approach can adequately address the strong endogeneity, multicollinearity, and model misspecification incorporated in the data-generating process. In particular, the unregularized Lee estimator fails because instability in the first-stage multinomial logit propagates into the second-stage correction term, producing unreliable estimates.

By contrast, the regularized estimators yield substantial improvements in both bias and MSE. The Ridge-based correction reduces the bias to approximately 40–41 and lowers the MSE by more than 90%, to around 1,600. Although still biased, this result highlights the stabilizing effect of penalization. The Lasso-based correction performs notably better, with mean bias close to zero (approximately −1.8 to −2.0) and an MSE of only about 3–4. This represents both low bias and low variance performance. The superior behavior of the Lasso estimator arises from its ability to identify and retain the sparse set of relevant covariates in the selection equation, leading to more accurate estimates of the choice probabilities and, consequently, of the selection correction term. Notably, the relative performance of the regularized estimators remains stable as dimensionality increases, even when $p > n$, underscoring their suitability for high-dimensional empirical applications.

Table 1: Monte Carlo results for the linear effect ($\beta_{true} = 2$)

	Bias				MeanSquaredError			
Scenario	Naive	Lee	Ridge	Lasso	Naive	Lee	Ridge	Lasso
$p = 5$	144.87	145.65	41.10	-1.96	21009.60	21236.81	1690.62	3.86
$p = 100$	141.88	141.87	40.09	-1.82	20157.16	20155.75	1608.97	3.32
$p = 1000$	141.80	141.82	40.04	-1.81	20135.62	20142.97	1605.34	3.27

Notes: Results averaged over 500 replications with $n = 500$. The DGP features strong sparsity and multicollinearity in Z, nonlinear selection and outcome equations, heteroskedastic and non-normal errors, and endogeneity induced by a common factor. Naive denotes OLS on the selected subsample with y regressed on X only. Lee refers to the unregularized two-step procedure. Ridge and Lasso apply penalized estimation in both stages, leaving the correction term unpenalized in the second stage.

4.2 Empirical Validation

To demonstrate the empirical applicability of the proposed method, it is applied to the `ModeCanada` dataset available in the `mlogit` package for R (Croissant, 2020). The dataset contains revealed preference data on transportation mode choice between Montreal and Toronto. The alternatives include air, bus, car, and train. Explanatory variables encompass alternative-specific attributes such as cost and time, as well as individual-specific income.

The exercise is structured as an illustrative demonstration of bias correction rather than a substantive empirical study. The analysis focuses on individuals selecting the "car" mode. A synthetic outcome variable, y, representing "monthly car-related expenses," is generated and observed only for those who chose the car. The data-generating process for y is designed to mirror the complexities in the simulation: the true linear effect of income on expenses is set to $\beta = 0.8$, and the model includes nonlinear components and a source of endogeneity. Specifically, a latent factor representing an individual's "propensity for car travel" influences both the mode choice and the outcome, creating correlation between selection and outcome disturbances. The sample consists of 2,779 observations, of which 1,138 (41.0%) correspond to car users.

Table 2 presents the estimates of the income effect on monthly car expenses. The Naive OLS and unregularized Lee estimators perform poorly, producing inflated estimates of β (23.95 and 23.85, respectively), far above the true value of 0.8. These results illustrate the inability of conventional estimators to correct for endogeneity and selection bias in this setting. The Ridge-based correction substantially reduces the bias, yielding an estimate of 11.96. Although still upwardly biased, it captures a significant portion of the true relationship. The Lasso-based correction provides the most accurate result, with an estimated coefficient of 1.28—close to the true value—and a marked reduction in the in-sample root mean squared error (RMSE). Specifically, the Lasso model achieves an RMSE of 16.20, compared to more than 120 for the unregularized approaches. These findings confirm the practical advantages of applying regularization in both stages of selection correction under complex, high-dimensional conditions.

Table 2: Empirical Results on `ModeCanada` Dataset ($\beta_{true} = 0.8$)

Method	Estimated β (Income)	In-sample RMSE
Naive OLS	23.9474	123.9620
Lee (unregularized)	23.8538	123.3901
Regularized Lee (Ridge)	11.9551	80.2575
Regularized Lee (LASSO)	1.2813	16.1992

Notes: Results are based on the `ModeCanada` dataset with a constructed outcome variable y (monthly car expenses) observed only for individuals selecting "car". The DGP includes strong endogeneity, nonlinearities, and heteroskedasticity. The true linear effect of income is $\beta = 0.8$. Naive OLS ignores selection; Lee applies the unregularized two-step correction. The regularized versions employ penalized estimation in both stages.

5 Conclusion

This note presents a regularized two-stage correction framework for addressing selection bias in multinomial choice models under high-dimensional settings. By combining penalized estimation in both the selection and outcome equations while leaving the correction term unpenalized, the approach stabilizes estimation in the presence of multicollinearity and sparsity. Simulation evidence indicates that regularization substantially improves finite-sample performance, and an empirical illustration demonstrates its usefulness in mitigating selection-related bias. The framework provides a computationally straightforward extension of classical correction methods that remains applicable in modern data-rich environments. These results suggest that incorporating regularization into selection correction offers a practical and robust tool for applied econometric analysis.

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Appendix

A Monte Carlo Design

The finite-sample properties of the proposed method are evaluated through Monte Carlo experiments. The simulation design is intentionally challenging, incorporating several features frequently encountered in empirical data that violate standard assumptions. The sample size is fixed at $n = 500$, while the number of covariates p in the selection equation varies from 5 to 1000, covering both conventional and high-dimensional ($p > n$) regimes.

The data-generating process (DGP) possesses the following characteristics. A latent common factor w induces strong endogeneity by jointly affecting the selection utilities, the main regressor X, and the outcome error u. The covariates Z exhibit substantial multicollinearity, with pairwise correlations of approximately 0.9. The selection mechanism follows a three-alternative ($J = 3$) multinomial logit model. The utilities are nonlinear functions of Z and are influenced by w, generating correlation between selection and outcome unobservables. The true coefficient vectors in the selection equation are sparse, implying that only a small subset of covariates has a nonzero effect—an environment in which the LASSO penalty is expected to perform well.

The regressor of interest, X, is defined as a function of selected components of Z and the latent factor w, combined with additional nonlinear terms and measurement error, resulting in endogeneity. The outcome disturbance u follows a standardized, non-normal distribution (e.g., Gamma) and is heteroskedastic, with its variance increasing in X. The true outcome model, observed only when $D_i = 1$, is nonlinear:

$$y_i = 2X_i + 0.3X_i^2 + 0.2\sin(X_i) + u_i. \tag{8}$$

Thus, the true linear effect of interest is $\beta = 2$.

Four estimators are compared across 500 replications for each configuration:

1. *Naive OLS:* A simple regression of y on X using only the selected subsample, ignoring both selection bias and nonlinearities.

2. *Lee (unregularized):* The classical two-step estimator using an unregularized multinomial logit in the first stage and OLS on y, X, and $\hat{\lambda}$ in the second.

3. *Regularized Lee (Ridge):* The proposed method with ridge regression ($\alpha = 0$) applied in both stages.

4. *Regularized Lee (LASSO):* The proposed method with lasso regression ($\alpha = 1$) applied in both stages.

The estimators' performance is evaluated in terms of the bias and mean squared error (MSE) of $\hat{\beta}$ relative to the true value. This design facilitates a transparent assessment of the relative gains from introducing regularization into both stages of the selection correction process.

B Algorithmic Summary of the Estimation Procedure

Algorithm 1 Regularized Two-Stage Correction for Multinomial Selection

Require: Observed data $\{y_i, X_i, Z_i, D_i\}_{i=1}^n$, where y_i is observed only if $D_i = 1$. Regularization parameter type $\alpha \in \{0, 1\}$ (0 = Ridge, 1 = LASSO).

procedure $\{y_i, X_i, Z_i, D_i\}, \alpha($
$)$

Stage 1: Regularized Selection Model

Estimate a penalized multinomial logit of D_i on Z_i via cross-validated `glmnet`:

$$\max_{\{\gamma_j\}} \left\{ \sum_{i=1}^{n} \log L(\{\gamma_j\}|D_i, Z_i) - \lambda_1 \sum_{j=1}^{J-1} \left[(1-\alpha)\|\gamma_j\|_2^2/2 + \alpha\|\gamma_j\|_1\right] \right\}.$$

Obtain fitted choice probabilities $\hat{P}_{i1} = \Pr(D_i = 1|Z_i)$.

Compute the Lee-type correction term for each observation:

$$\hat{\lambda}_i = -\frac{\phi\big(\Phi^{-1}(\hat{P}_{i1})\big)}{\hat{P}_{i1}}.$$

Stage 2: Regularized Outcome Model

Restrict to the selected subsample $\mathcal{S} = \{i \mid D_i = 1\}$.

Construct the augmented regressor matrix:

$$X_i^* = [X_i,\, X_i^2,\, X_i^3,\, \hat{\lambda}_i].$$

Estimate a penalized linear regression on $\mathcal{S}$:

$$\min_{\beta,\delta,\theta} \sum_{i \in \mathcal{S}} \left(y_i - X_i^{*\top}[\beta, \delta, \theta]\right)^2 + \lambda_2\left[(1-\alpha)\|\beta, \delta\|_2^2/2 + \alpha\|\beta, \delta\|_1\right].$$

The correction term coefficient θ is excluded from penalization (`penalty.factor=0`).

Extract the estimated coefficient $\hat{\beta}$ corresponding to the linear term X_i.

return $\hat{\beta}$.

end procedure